Shahar Kvatinsky

associate professor

基本信息

姓名： Shahar Kvatinsky　**所属院校：** 以色列理工学院

邮箱： morgenshtein@technion.ac.il　**地区：** Israel

年龄范围： 40岁至50岁　**企业经历：** 有

学科领域

电子与计算机工程　微电子科学与工程　集成电路设计与集成系统

研究领域

计算机体系结构　微电子器件与制造　存储器设计与架构　低功耗设计与能耗优化

纳米电子与先进工艺

个人简介

Shahar Kvatinsky is an Associate Professor at the Andrew and Erna Viterbi Faculty of Electrical and Computer Engineering at Technion – Israel Institute of Technology. His research focuses on circuit design, computer architecture, artificial intelligence, VLSI systems, and energy efficiency in computers. Kvatinsky leads the Advanced Circuits Research Center at Technion and is the chair of the IEEE Circuits and Systems in Israel. He has received numerous awards for his research and teaching, including the 2021 Norman Seiden Prize for Academic Excellence and multiple Hershel Rich Technion Innovation Awards. His work emphasizes emerging memory technologies and energy-efficient architectures.

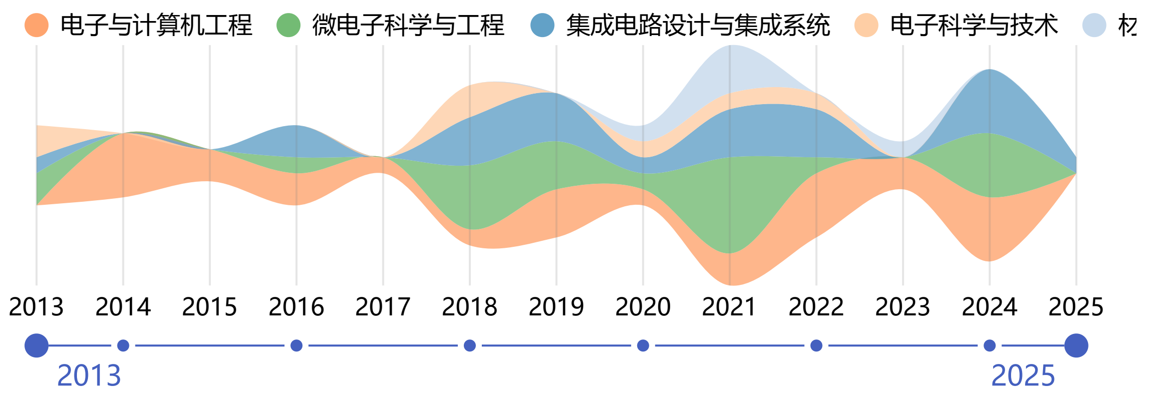
工作经历

Technion - Israel Institute of Technology 2015-09-01 -

教育经历

Technion Israel Institute of Technology - PhD - 2014-09

近年发表的论文



IMPACT: In-Memory ComPuting Architecture based on Y-FlAsh Technology for Coalesced Tsetlin machine inference

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES

Ghazal, Omar; Wang, Wei; Kvatinsky, Shahar; Merchant, Farhad; Yakovlev, Alex; Shafik, Rishad

被引用次数：0 2025

Assessing the Performance of Stateful Logic in 1-Selector-1-RRAM Crossbar Arrays

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Tyagi, Arjun; Kvatinsky, Shahar

被引用次数：1 2024

Bitwise Logic Using Phase Change Memory Devices Based on the Pinatubo Architecture

PROCEEDINGS OF THE 37TH INTERNATIONAL CONFERENCE ON VLSI DESIGN, VLSID 2024 AND 23RD INTERNATIONAL CONFERENCE ON EMBEDDED SYSTEMS, ES 2024

Aflalo, Noa; Yalon, Eilam; Kvatinsky, Shahar

被引用次数：1 2024

Abstract Thinking of Beginning Electrical Engineering and Computer Science Students

TOWARDS A HYBRID, FLEXIBLE AND SOCIALLY ENGAGED HIGHER EDUCATION, VOL 3, ICL 2023

Gero, Aharon; Hadish, Mohammed Ali; Kvatinsky, Shahar

被引用次数：0 2024

A Concealable RRAM Physical Unclonable Function Compatible with In-Memory Computing

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Li, Jiang; Cui, Yijun; Wang, Chenghua; Liu, Weiqiang; Kvatinsky, Shahar

被引用次数：0 2024

Asymmetric and Symmetric Single-Pole Double-Throw With Improved Power Handling Using Indirectly Heated Phase-Change Switches

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wainstein, Nicolas; Orren, Ami; Nir-Harwood, Rivka-Galya; Yalon, Eilam; Kvatinsky, Shahar

被引用次数：0 2024

Roadmap to neuromorphic computing with emerging technologies

APL MATERIALS

Mehonic, Adnan; Ielmini, Daniele; Roy, Kaushik; Mutlu, Onur; Kvatinsky, Shahar; Serrano-Gotarredona, Teresa; Linares-Barranco, Bernabe; Spiga, Sabina; Savel'ev, Sergey; Balanov, Alexander G.; Chawla, Nitin; Desoli, Giuseppe; Malavena, Gerardo; Monzio Compagnoni, Christian; Wang, Zhongrui; Yang, J. Joshua; Sarwat, Syed Ghazi; Sebastian, Abu; Mikolajick, Thomas; Slesazeck, Stefan; Noheda, Beatriz; Dieny, Bernard; Hou, Tuo-Hung (Alex); Varri, Akhil; Brueckerhoff-Plueckelmann, Frank; Pernice, Wolfram; Zhang, Xixiang; Pazos, Sebastian; Lanza, Mario; Wiefels, Stefan; Dittmann, Regina; Ng, Wing H.; Buckwell, Mark; Cox, Horatio R. J.; Mannion, Daniel J.; Kenyon, Anthony J.; Lu, Yingming; Yang, Yuchao; Querlioz, Damien; Hutin, Louis; Vianello, Elisa; Chowdhury, Sayeed Shafayet; Mannocci, Piergiulio; Cai, Yimao; Sun, Zhong; Pedretti, Giacomo; Strachan, John Paul; Strukov, Dmitri; Le Gallo, Manuel; Ambrogio, Stefano; Valov, Ilia; Waser, Rainer

被引用次数：0 2024

Experimental Demonstration of Non-Stateful In-Memory Logic With 1T1R OxRAM Valence Change Mechanism Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Padberg, Henriette; Regev, Amir; Piccolboni, Giuseppe; Bricalli, Alessandro; Molas, Gabriel; Nodin, Jean Francois; Kvatinsky, Shahar

被引用次数：1 2024

Understanding Bulk-Bitwise Processing In-Memory Through Database Analytics

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Perach, Ben; Ronen, Ronny; Kimelfeld, Benny; Kvatinsky, Shahar

被引用次数：0 2024

TDPP: 2-D Permutation-Based Protection of Memristive Deep Neural Networks

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zou, Minhui; Zhu, Zhenhua; Greenberg-Toledo, Tzofnat; Leitersdorf, Orian; Li, Jiang; Zhou, Junlong; Wang, Yu; Du, Nan; Kvatinsky, Shahar

被引用次数：0 2024

A full spectrum of computing-in-memory technologies

NATURE ELECTRONICS

Sun, Zhong; Kvatinsky, Shahar; Si, Xin; Mehonic, Adnan; Cai, Yimao; Huang, Ru

被引用次数：10 2023

ClaPIM: Scalable Sequence Classification Using Processing-in-Memory

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Khalifa, Marcel; Hoffer, Barak; Leitersdorf, Orian; Hanhan, Robert; Perach, Ben; Yavits, Leonid; Kvatinsky, Shahar

被引用次数：1 2023

Scalable Al2O3-TiO2 Conductive Oxide Interfaces as Defect Reservoirs for Resistive Switching Devices

ADVANCED ELECTRONIC MATERIALS

Li, Yang; Wang, Wei; Zhang, Di; Baskin, Maria; Chen, Aiping; Kvatinsky, Shahar; Yalon, Eilam; Kornblum, Lior

被引用次数：8 2023

Enhancing Security of Memristor Computing System Through Secure Weight Mapping

2022 IEEE COMPUTER SOCIETY ANNUAL SYMPOSIUM ON VLSI (ISVLSI 2022)

Zou, Minhui; Zhou, Junlong; Cui, Xiaotong; Wang, Wei; Kvatinsky, Shahar

被引用次数：5 2022

MatPIM: Accelerating Matrix Operations with Memristive Stateful Logic

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Leitersdorf, Orian; Ronen, Ronny; Kvatinsky, Shahar

被引用次数：1 2022

Stateful Logic Using Phase Change Memory

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Hoffer, Barak; Wainstein, Nicolas; Neumann, Christopher M. M.; Pop, Eric; Yalon, Eilam; Kvatinsky, Shahar

被引用次数：5 2022

Efficient Training of the Memristive Deep Belief Net Immune to Non-Idealities of the Synaptic Devices

ADVANCED INTELLIGENT SYSTEMS

Wang, Wei; Hoffer, Barak; Greenberg-Toledo, Tzofnat; Li, Yang; Zou, Minhui; Herbelin, Eric; Ronen, Ronny; Xu, Xiaoxin; Zhao, Yulin; Yang, Jianguo; Kvatinsky, Shahar

被引用次数：8 2022

A memristive deep belief neural network based on silicon synapses

NATURE ELECTRONICS

Wang, Wei; Danial, Loai; Li, Yang; Herbelin, Eric; Pikhay, Evgeny; Roizin, Yakov; Hoffer, Barak; Wang, Zhongrui; Kvatinsky, Shahar

被引用次数：22 2022

Superconductive Logic Using 2φ-Josephson Junctions With Half Flux Quantum Pulses

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Salameh, Issa; Friedman, Eby G.; Kvatinsky, Shahar

被引用次数：11 2022

MultPIM: Fast Stateful Multiplication for Processing-in-Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Leitersdorf, Orian; Ronen, Ronny; Kvatinsky, Shahar

被引用次数：13 2022

The Bitlet Model: A Parameterized Analytical Model to Compare PIM and CPU Systems

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Ronen, Ronny; Eliahu, Adi; Leitersdorf, Orian; Peled, Natan; Korgaonkar, Kunal; Chattopadhyay, Anupam; Ben Perach; Kvatinsky, Shahar

被引用次数：13 2022

C-AND: Mixed Writing Scheme for Disturb Reduction in 1T Ferroelectric FET Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Dahan, Mor M.; Breyer, Evelyn T.; Slesazeck, Stefan; Mikolajick, Thomas; Kvatinsky, Shahar

被引用次数：8 2022

Uncovering Phase Change Memory Energy Limits by Sub-Nanosecond Probing of Power Dissipation Dynamics

ADVANCED ELECTRONIC MATERIALS

Stern, Keren; Wainstein, Nicolas; Keller, Yair; Neumann, Christopher M.; Pop, Eric; Kvatinsky, Shahar; Yalon, Eilam

被引用次数：16 2021

Harnessing Conductive Oxide Interfaces for Resistive Random-Access Memories

FRONTIERS IN PHYSICS

Li, Yang; Kvatinsky, Shahar; Kornblum, Lior

被引用次数：6 2021

Sub-Nanosecond Pulses Enable Partial Reset for Analog Phase Change Memory

IEEE ELECTRON DEVICE LETTERS

Stern, Keren; Wainstein, Nicolas; Keller, Yair; Neumann, Christopher M.; Pop, Eric; Kvatinsky, Shahar; Yalon, Eilam

被引用次数：10 2021

Radiofrequency Switches Based on Emerging Resistive Memory Technologies-A Survey

PROCEEDINGS OF THE IEEE

Wainstein, Nicolas; Adam, Gina; Yalon, Eilam; Kvatinsky, Shahar

被引用次数：35 2021

Comments on Experimental Demonstration of Memristor-Aided Logic (MAGIC) Using Valence Change Memory (VCM)

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hoffer, Barak; Rana, Vikas; Menzel, Stephan; Waser, Rainer; Kvatinsky, Shahar

被引用次数：0 2021

Indirectly Heated Switch as a Platform for Nanosecond Probing of Phase Transition Properties in Chalcogenides

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wainstein, Nicolas; Ankonina, Guy; Swoboda, Timm; Rojo, Miguel Munoz; Kvatinsky, Shahar; Yalon, Eilam

被引用次数：8 2021

Training of quantized deep neural networks using a magnetic tunnel junction-based synapse

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Greenberg-Toledo, Tzofnat; Perach, Ben; Hubara, Itay; Soudry, Daniel; Kvatinsky, Shahar

被引用次数：2 2021

Physical based compact model of Y-Flash memristor for neuromorphic computation

APPLIED PHYSICS LETTERS

Wang, Wei; Danial, Loai; Herbelin, Eric; Hoffer, Barak; Oved, Batel; Greenberg-Toledo, Tzofnat; Pikhay, Evgeny; Roizin, Yakov; Kvatinsky, Shahar

被引用次数：9 2021

Improving Efficiency and Lifetime of Logic-in-Memory by Combining IMPLY andMAGIC Families

JOURNAL OF SYSTEMS ARCHITECTURE

Zou, Minhui; Zhou, Junlong; Sun, Jin; Wang, Chengliang; Kvatinsky, Shahar

被引用次数：2 2021

(V)TEAM for SPICE Simulation of Memristive Devices With Improved Numerical Performance

IEEE ACCESS

Biolek, Dalibor; Kolka, Zdenek; Biolkova, Viera; Biolek, Zdenek; Kvatinsky, Shahar

被引用次数：12 2021

multiPULPly: A Multiplication Engine for Accelerating Neural Networks on Ultra-low-power Architectures

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Eliahu, Adi; Ronen, Ronny; Gaillardon, Pierre-Emmanuel; Kvatinsky, Shahar

被引用次数：5 2021

SIMPLER MAGIC: Synthesis and Mapping of In-Memory Logic Executed in a Single Row to Improve Throughput

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ben-Hur, Rotem; Ronen, Ronny; Haj-Ali, Ameer; Bhattacharjee, Debjyoti; Eliahu, Adi; Peled, Natan; Kvatinsky, Shahar

被引用次数：67 2020

Experimental Demonstration of Memristor-Aided Logic (MAGIC) Using Valence Change Memory (VCM)

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hoffer, Barak; Rana, Vikas; Menzel, Stephan; Waser, Rainer; Kvatinsky, Shahar

被引用次数：57 2020

Cytomorphic Electronics With Memristors for Modeling Fundamental Genetic Circuits

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Hanna, Hanna Abo; Danial, Loai; Kvatinsky, Shahar; Daniel, Ramez

被引用次数：6 2020

Compact Modeling and Electrothermal Measurements of Indirectly Heated Phase-Change RF Switches

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wainstein, Nicolas; Ankonina, Guy; Kvatinsky, Shahar; Yalon, Eilam

被引用次数：11 2020

Oxide 2D electron gases as a reservoir of defects for resistive switching

APPLIED PHYSICS LETTERS

Miron, Dror; Cohen-Azarzar, Dana; Hoffer, Barak; Baskin, Maria; Kvatinsky, Shahar; Yalon, Eilam; Kornblum, Lior

被引用次数：11 2020

Understanding the influence of device, circuit and environmental variations on real processing in memristive memory using Memristor Aided Logic

MICROELECTRONICS JOURNAL

Wald, Nimrod; Kvatinsky, Shahar

被引用次数：21 2019

Adaptive programming in multi-level cell ReRAM

MICROELECTRONICS JOURNAL

Ramadan, Misbah; Wainstein, Nicolas; Ginosar, Ran; Kvatinsky, Shahar

被引用次数：11 2019

Two-terminal floating-gate transistors with a low-power memristive operation mode for analogue neuromorphic computing

NATURE ELECTRONICS

Danial, Loai; Pikhay, Evgeny; Herbelin, Eric; Wainstein, Nicolas; Gupta, Vasu; Wald, Nimrod; Roizin, Yakov; Daniel, Ramez; Kvatinsky, Shahar

被引用次数：107 2019

An Asynchronous and Low-Power True Random Number Generator Using STT-MTJ

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Perach, Ben; Kvatinsky, Shahar

被引用次数：21 2019

Supporting the Momentum Training Algorithm Using a Memristor-Based Synapse

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Greenberg-Toledo, Tzofnat; Mazor, Roee; Haj-Ali, Ameer; Kvatinsky, Shahar

被引用次数：16 2019

CONCEPT: A Column-Oriented Memory Controller for Efficient Memory and PIM Operations in RRAM

IEEE MICRO

Talati, Nishil; Ha, Heonjae; Perach, Ben; Ronen, Ronny; Kvatinsky, Shahar

被引用次数：13 2019

Analysis of the row grounding technique in a memristor-based crossbar array

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Dozortsev, Alexander; Goldshtein, Israel; Kvatinsky, Shahar

被引用次数：20 2018

A Lumped RF Model for Nanoscale Memristive Devices and Nonvolatile Single-Pole Double-Throw Switches

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Wainstein, Nicolas; Kvatinsky, Shahar

被引用次数：6 2018

Not in Name Alone: A Memristive Memory Processing Unit for Real In-Memory Processing

IEEE MICRO

Haj-Ali, Ameer; Ben-Hur, Rotem; Wald, Nimrod; Ronen, Ronny; Kvatinsky, Shahar

被引用次数：28 2018

Breaking Through the Speed-Power-Accuracy Tradeoff in ADCs Using a Memristive Neuromorphic Architecture

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTATIONAL INTELLIGENCE

Danial, Loai; Wainstein, Nicolas; Kraus, Shraga; Kvatinsky, Shahar

被引用次数：46 2018

DIDACTIC: A Data-Intelligent Digital-to-Analog Converter with a Trainable Integrated Circuit using Memristors

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Danial, Loai; Wainstein, Nicols; Kraus, Shraga; Kvatinsky, Shahar

被引用次数：17 2018

IMAGING: In-Memory AlGorithms for Image processiNG

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Haj-Ali, Ameer; Ben-Hur, Rotem; Wald, Nimrod; Ronen, Ronny; Kvatinsky, Shahar

被引用次数：36 2018

TIME-Tunable Inductors Using MEmristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wainstein, Nicolas; Kvatinsky, Shahar

被引用次数：16 2018

Dark Memory and Accelerator-Rich System Optimization in the Dark Silicon Era

IEEE DESIGN & TEST

Pedram, Ardavan; Richardson, Stephen; Horowitz, Mark; Galal, Sameh; Kvatinsky, Shahar

被引用次数：90 2017

Logic Design Within Memristive Memories Using Memristor-Aided loGIC (MAGIC)

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Talati, Nishil; Gupta, Saransh; Mane, Pravin; Kvatinsky, Shahar

被引用次数：253 2016

Resistive GP-SIMD Processing-In-Memory

ACM TRANSACTIONS ON ARCHITECTURE AND CODE OPTIMIZATION

Morad, Amir; Yavits, Leonid; Kvatinsky, Shahar; Ginosar, Ran

被引用次数：30 2016

Information-Theoretic Sneak-Path Mitigation in Memristor Crossbar Arrays

IEEE TRANSACTIONS ON INFORMATION THEORY

Cassuto, Yuval; Kvatinsky, Shahar; Yaakobi, Eitan

被引用次数：45 2016

Multistate Register Based on Resistive RAM

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Patel, Ravi; Kvatinsky, Shahar; Friedman, Eby G.; Kolodny, Avinoam

被引用次数：8 2015

Resistive Associative Processor

IEEE COMPUTER ARCHITECTURE LETTERS

Yavits, Leonid; Kvatinsky, Shahar; Morad, Amir; Ginosar, Ran

被引用次数：66 2015

Logic operations in memory using a memristive Akers array

MICROELECTRONICS JOURNAL

Levy, Yifat; Bruck, Jehoshua; Cassuto, Yuval; Friedman, Eby G.; Kolodny, Avinoam; Yaakobi, Eitan; Kvatinsky, Shahar

被引用次数：67 2014

Memristor-Based Material Implication (IMPLY) Logic: Design Principles and Methodologies

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kvatinsky, Shahar; Satat, Guy; Wald, Nimrod; Friedman, Eby G.; Kolodny, Avinoam; Weiser, Uri C.

被引用次数：448 2014

MAGIC-Memristor-Aided Logic

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kvatinsky, Shahar; Belousov, Dmitry; Liman, Slavik; Satat, Guy; Wald, Nimrod; Friedman, Eby G.; Kolodny, Avinoam; Weiser, Uri C.

被引用次数：617 2014

Memristor-Based Multithreading

IEEE COMPUTER ARCHITECTURE LETTERS

Kvatinsky, Shahar; Nacson, Yuval H.; Etsion, Yoav; Friedman, Eby G.; Kolodny, Avinoam; Weiser, Uri C.

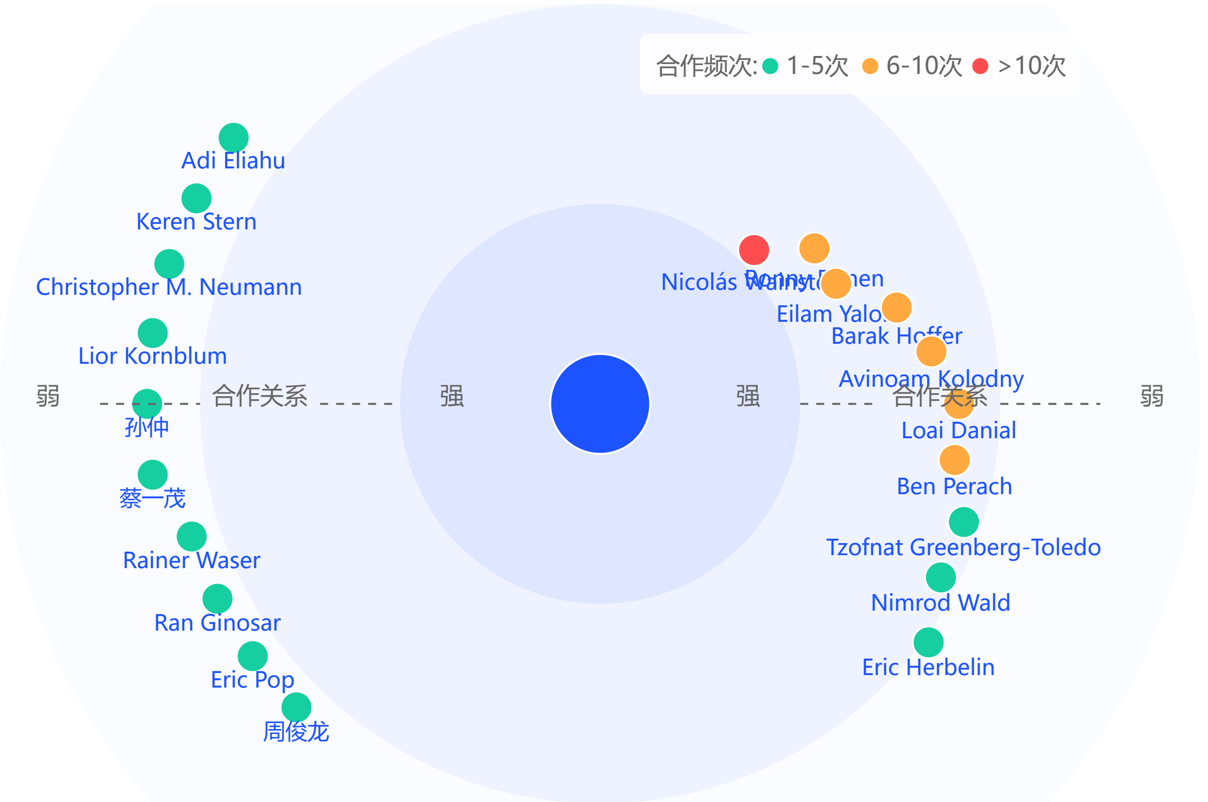
被引用次数：15 2014

TEAM: ThrEshold Adaptive Memristor Model

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS
Kvatinsky, Shahar; Friedman, Eby G.; Kolodny, Avinoam; Weiser, Uri C.
被引用次数：602 2013

The Desired Memristor for Circuit Designers
IEEE CIRCUITS AND SYSTEMS MAGAZINE
Kvatinsky, Shahar; Friedman, Eby G.; Kolodny, Avinoam; Weiser, Uri C.
被引用次数：69 2013

学术合作



Nicolás Wainstein 合作次数：12
以色列理工学院

Ronny Ronen 合作次数：10
以色列理工学院

Eilam Yalon 合作次数：10
以色列理工学院

Barak Hoffer 合作次数：8
以色列理工学院

Avinoam Kolodny 合作次数：7
以色列理工学院

还有 49 个合作者